

Accessibility as a Service (AaaS) Infrastructure for Public Institutions

Problem Statement Title- **Lack of Inclusive Digital Accessibility Infrastructure in Public Institutions**

Idea Title- **Accessibility as a Service (AaaS) Infrastructure for Public Institutions**

Theme- **Accessibility, Ecommerce & Cyber Security**

PS Category- Software

Team ID-

Team Name- **SUBARNAREKHA**

Accessibility as a Service (AaaS) Infrastructure for Public Institutions

➤ Accessibility as a Service (AaaS) is a centralized, cloud-based infrastructure designed to make public institutions digitally inclusive for persons with disabilities and diverse linguistic backgrounds. The system provides essential accessibility services such as speech-to-text, text-to-speech with screen readers, learning-disability–friendly content formats, accessible examination systems, and multi-language including local language support. Instead of each institution developing separate accessibility solutions, this platform offers plug-and-play accessibility APIs and interfaces that can be integrated into websites, applications, kiosks, classrooms, and examination systems, ensuring equal access to information and services for all citizens.

➤ **How it addresses the problem-**

Despite digitalization in public institutions such as schools, universities, government offices, and examination boards, **accessibility remains severely inadequate.**

Key problems include: 1. Lack of support for visually, hearing, and learning-impaired users 2. Absence of screen readers and assistive interfaces on official portals 3. Inaccessible digital examinations and learning materials

4. Limited availability of local and regional language support

5. High cost and complexity of implementing accessibility solutions individually

As a result, millions of people are excluded from education, employment, and public services.

➤ **Innovation and uniqueness of the solution**

- ✓ **Centralized Accessibility as a Service model** (not institution-specific)
- ✓ **Reusable and scalable APIs** instead of custom implementations
- ✓ Supports **learning disabilities**, which are often ignored
- ✓ Integrates **accessible exams** (extra time, audio support, dyslexia-friendly fonts)
- ✓ Emphasizes **local Indian languages**, not just English
- ✓ Cost-effective and easy to deploy for public institutions
- ✓ Works across **web, mobile, and physical kiosks**

This approach reduces duplication, cost, and technical barriers while ensuring nationwide accessibility compliance.

TECHNICAL APPROACH

Software Components:

- Speech-to-Text Module
- Text-to-Speech & Screen Reader Engine
- Accessibility API Gateway
- Language Translation Engine
- Accessible Examination Module
- Admin Dashboard

Hardware (Optional):

- Microphone and speakers
- Accessibility kiosks (touch + voice enabled)
- Low-cost tablets for classrooms
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

Implementation-

- Requirement Analysis from public institutions
- Accessibility standards mapping (WCAG)
- Development of core accessibility services
- API-based integration with institutional platforms
- Testing with real users (PwDs & educators)
- Deployment & training

FEASIBILITY AND VIABILITY

- **Technically feasible** using existing AI and cloud technologies
- **Economically viable** due to shared infrastructure
- **Socially relevant** and aligned with Digital India & Inclusive India goals
- Can be scaled gradually from pilot institutions to nationwide adoption

Potential Challenges and Risks

- . Accuracy in speech recognition for local languages
- . Resistance to adoption in traditional institutions
- . Data privacy and security concerns
- . Accessibility awareness among staff

Strategies for Overcoming These Challenges

- Training AI models using local language datasets
- Awareness and training programs for institutions
- Strong encryption and compliance with data protection laws
- Government policy alignment and incentives

IMPACT AND BENEFITS

Potential Impact on the Target Audience

- Equal access to education and government services
- Increased participation of persons with disabilities
- Improved learning outcomes
- Reduced digital divide
- Empowerment through inclusive technology

Beneficiary	Benefit
Students with disabilities	Inclusive learning & exams
Public institutions	Cost-effective accessibility
Government	Compliance & social inclusion
Citizens	Equal digital access

THANK YOU